

# TAVAHEED TARIQ

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## EXPERIENCE

### Web Development Team Lead

Gaash Lab NIT Srinagar

July 2024 – Current  
Srinagar, India – On-Site

- Leading a 15+ member team in building full-stack web applications for institutional and research-based projects.
- Designed and implemented scalable architectures integrating React (frontend) and Django/Flask (backend) for production-level deployment.
- Established CI/CD pipelines using Docker and GitHub Actions to ensure reliable deployment and version control.
- Mentored junior developers on web frameworks, REST API design, and best practices in UI/UX development.

### Project Intern

Indian Institute of Technology (IIT) Delhi

Dec 2024 – May 2025  
Delhi, India – Remote

- Developed and fine-tuned an Urdu OCR system based on the PARSeq transformer architecture for scene text recognition.
- Processed, cleaned, and labeled over 28 million images to create a robust dataset for training and validation.
- Achieved State-of-the-Art (SOTA) accuracy in both Character Recognition Rate (CRR) and Word Recognition Rate (WRR) on benchmark test sets.
- Documented model design, evaluation metrics, and dataset preprocessing for reproducibility and research dissemination.

### Research Intern

Indian Institute of Technology (IIT) Ropar

Dec 2024 – Feb 2025  
Punjab, India – On-Site

- Conducted research on predicting initial and post-cycle charge capacity of lithium-ion batteries using ML-based regression models.
- Preprocessed and analyzed a dataset of 168 samples covering 20 dopant types with physical and chemical descriptors.
- Implemented and compared models including Random Forest, SVM, Ridge, and XGBoost for performance benchmarking.
- Identified XGBoost as the best-performing model with high accuracy and generalization ability.

## PROJECTS

### WP-Scanner

[GitHub](#)

- Designed and developed a full-stack web platform for automated penetration testing of WordPress websites using **React**, **Flask**, and **MongoDB**.
- Implemented modules for **plugin/theme vulnerability scanning**, brute-force testing, and outdated component detection.
- Built a dynamic reporting dashboard to visualize scan summaries, security grades, and mitigation suggestions.
- Deployed Docker-based microservices for scalability and isolation of the scanning engine.

### NIT Datesheet Generator

- Created an intelligent scheduling tool for **automatic generation of exam datesheets** for all engineering courses at NIT Srinagar.
- Designed constraint-based algorithms to ensure **conflict-free exam allocation** across departments and semesters.
- Built a user-friendly interface for administrators to manage subjects, departments, and exam slots dynamically.
- Exported results in tabular and PDF formats for official use, saving significant manual scheduling effort.

### AffectSense

[GitHub](#)

- Developed a deep learning-based **emotion recognition system** using a custom **ResNet50-based CNN architecture**.
- Trained on benchmark emotion datasets (FER-2013, CK+) with extensive data augmentation for generalization.
- Achieved **82% classification accuracy**, outperforming baseline CNN models in facial emotion recognition.
- Deployed a lightweight inference pipeline supporting webcam-based real-time emotion detection.

## Kernel Hacking

[GitHub](#)

- Implemented multiple **custom system calls** in the Linux kernel to extend functionality for file handling and process tracing.
- Modified and rebuilt kernel source, exploring low-level modules for memory management, scheduling, and inter-process communication.
- Developed a user-space interface in C to invoke and test new system calls efficiently.
- Debugged kernel crashes using **GDB and printk logs**, enhancing understanding of OS internals and debugging workflows.
- Contributed to open-source documentation outlining the kernel compilation, patching, and testing process.

## TECHNICAL SKILLS

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**Languages:** C, C++, Python, JavaScript

**Frameworks:** React, Flask, Node.js

**Databases:** MongoDB, SQL

**Tools:** Docker, Git, Linux, Postman

**Core:** DSA, OOP, OS, DBMS

**Other:** Machine Learning, Computer Vision

## PUBLICATIONS

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### QTrack: Query-Driven Reasoning for Multi-modal MOT

Tajamul Ashraf, **Tavaheed Tariq**, Sonia Yadav, Abrar Ul Riyaz, Wasif Tak, Moloud Abdar, Janibul Bashir  
arXiv preprint, 2026

[Webpage](#)

### GroundedSurg: A Multi-Procedure Benchmark for Language-conditioned Surgical Tool Segmentation

Tajamul Ashraf, Abrar Ul Riyaz, Wasif Tak, **Tavaheed Tariq**, Sonia Yadav, Moloud Abdar, Janibul Bashir  
arXiv preprint, 2026

[Link](#)

### GT: Context-Aware Grounded Teacher for Source-Free Object Detection

Tajamul Ashraf, Rajes Manna, Partha Sarathi Purkayastha, **Tavaheed Tariq**, Janibul Bashir  
arXiv preprint, 2025

[Link](#)

## EDUCATION

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**National Institute of Technology Srinagar**, Srinagar

Bachelor of Technology(B-Tech) Information Technology

Nov 2022 - present

CGPA: 9.06

**MPML Higher secondary**, Srinagar

HSC(10+2) PCM

Dec 2019 - Dec 2021

Percentage: 98.2%

## ACHIEVEMENTS

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- Achieved 1st place in "Bug Bounty" competition by efficiently identifying and solving bugs in provided programs.